Laborator
Fjalorë dhe Set

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Përmbajtja e laboratorit

1. Pako letrash
2. Fjalori i ditëlindjeve
3. Operacione me *set*
4. Moduli `pickle`
5. Eksport/Import `JSON`

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Qëllimi

- Të modelojmë një **pako standarde letrash** (52 letra) me **fjalor**.
- **Çelësat** (keys) janë emrat e letrave, p.sh. `'Queen of Hearts'`.
- **Vlerat** (values) janë vlerat numerike sipas Blackjack:
  - 2 - 10 marrin vlerën e vet.
  - `Jack`, `Queen`, `King` → 10.
  - `Ace` → 1 (në këtë program e thjeshtojmë si 1).
- Programi pyet sa letra të ndajë dhe shfaq totalin e vlerave.

Shënim

Kodi i mëposhtëm vendos **vlerën 1 për të gjitha Aset**. Më vonë mund ta zgjerojmë që Ace të jetë 1 ose 11 sipas zgjedhjes.

card_dealer.py main()

```python
import random

def main():
    deck = create_deck()

    num_cards = int(input('Sa letra te ndaj? '))

    deal_cards(deck, num_cards)

if __name__ == '__main__':
    main()
```

create_deck() kthen fjalorin me 52 letra I

```
def create_deck():
    deck = {
        'Ace of Spades':1, '2 of Spades':2, '3 of Spades':3,
        '4 of Spades':4, '5 of Spades':5, '6 of Spades':6,
        '7 of Spades':7, '8 of Spades':8, '9 of Spades':9,
        '10 of Spades':10, 'Jack of Spades':10,
        'Queen of Spades':10, 'King of Spades':10,

        'Ace of Hearts':1, '2 of Hearts':2, '3 of Hearts':3,
        '4 of Hearts':4, '5 of Hearts':5, '6 of Hearts':6,
        '7 of Hearts':7, '8 of Hearts':8, '9 of Hearts':9,
        '10 of Hearts':10, 'Jack of Hearts':10,
        'Queen of Hearts':10, 'King of Hearts':10,
```

create_deck() kthen fjalorin me 52 letra II

```
        'Ace of Clubs':1, '2 of Clubs':2, '3 of Clubs':3,
        '4 of Clubs':4, '5 of Clubs':5, '6 of Clubs':6,
        '7 of Clubs':7, '8 of Clubs':8, '9 of Clubs':9,
        '10 of Clubs':10, 'Jack of Clubs':10,
        'Queen of Clubs':10, 'King of Clubs':10,

        'Ace of Diamonds':1, '2 of Diamonds':2, '3 of Diamonds':3,
        '4 of Diamonds':4, '5 of Diamonds':5, '6 of Diamonds':6,
        '7 of Diamonds':7, '8 of Diamonds':8, '9 of Diamonds':9,
        '10 of Diamonds':10, 'Jack of Diamonds':10,
        'Queen of Diamonds':10, 'King of Diamonds':10
    }
    return deck
```

deal_cards(deck, number) ndarja e letrave

```
def deal_cards(deck, number):
    hand_value = 0

    if number > len(deck):
        number = len(deck)

    for _ in range(number):
        card = random.choice(list(deck))  # zgjedh nje kyç rastësor
        print(card)
        hand_value += deck[card]

    print(f'Vlera totale e dores: {hand_value}')
```

Çfarë rreziku kemi në këtë program?

Kujdes

Zgjedhja e kartës bëhet me `random.choice(list(deck))` dhe **nuk** e heq letrën nga pakoja; pra mund të bjerë e njëjta letër disa herë. Për të *mos* lejuar përsëritje, hiqe: `value = deck.pop(card)`.

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Qëllimi

- Ruaj emrat e miqve dhe ditëlindjet e tyre në një **fjalor**.
- Ofron një **menu** me zgjedhje:
  1. Kërko një ditëlindje
  2. Shto një të re
  3. Ndrysho një ekzistuese
  4. Fshi një hyrje
  5. Dil

birthdays.py konstantet dhe main() I

```
LOOK_UP = 1; ADD = 2; CHANGE = 3; DELETE = 4; QUIT = 5;
def main():
    birthdays = {}
    choice = 0

    while choice != QUIT:
        choice = get_menu_choice()

        if choice == LOOK_UP:
            look_up(birthdays)
        elif choice == ADD:
            add(birthdays)
        elif choice == CHANGE:
            change(birthdays)
        elif choice == DELETE:
            delete(birthdays)

if __name__ == '__main__':
    main()
```

birthdays.py konstantet dhe main() II

```python
def get_menu_choice():
    print()
    print('Friends and Their Birthdays')
    print('---------------------------')
    print('1. Look up a birthday')
    print('2. Add a new birthday')
    print('3. Change a birthday')
    print('4. Delete a birthday')
    print('5. Quit the program')
    print()

    choice = int(input('Enter your choice: '))
    while choice < LOOK_UP or choice > QUIT:
        choice = int(input('Enter a valid choice: '))
    return choice
```

birthdays.py konstantet dhe main() III

```
def look_up(birthdays):
    name = input('Enter a name: ')
    print(birthdays.get(name, 'Not found.'))

def add(birthdays):
    name = input('Enter a name: ')
    bday = input('Enter a birthday (p.sh. 5/7/1989): ')
    if name not in birthdays:
        birthdays[name] = bday
    else:
        print('That entry already exists.')
```

birthdays.py konstantet dhe main() IV

```
def change(birthdays):
    name = input('Enter a name: ')
    if name in birthdays:
        bday = input('Enter the new birthday: ')
        birthdays[name] = bday
    else:
        print('That name is not found.')

def delete(birthdays):
    name = input('Enter a name: ')
    if name in birthdays:
        del birthdays[name]
    else:
        print('That name is not found.')
```

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Qëllimi

Në këtë seksion shohim programin `sets.py`, i cili demonstron operacionet kryesore me `set`. Programi krijon dy **bashkësi**:

- `baseball` emrat e nxënësve në ekipin e baseball-it;
- `basketball` emrat e nxënësve në ekipin e basketball-it.

Pastaj programi kryen këto operacione:

- gjen **prèrjen** (`intersection`)
- gjen **bashkimin** (`union`)
- gjen **diferencën** `baseball - basketball`
- gjen **diferencën** `basketball - baseball`
- gjen **diferencën simetrike** (`symmetric_difference`)

sets.py I

```
baseball   = set(['Jodi', 'Carmen', 'Aida', 'Alicia'])
basketball = set(['Eva', 'Carmen', 'Alicia', 'Sarah'])

print('The following students are on the baseball team:')
for name in baseball: ]
    print(name)

print('\nThe following students are on the basketball team:')
for name in basketball:
    print(name)

print('\nThe following students play both baseball and basketball:')
for name in baseball.intersection(basketball):
    print(name)
```

sets.py II

```
print('\nThe following students play either baseball or basketball:')
for name in baseball.union(basketball):
    print(name)

print('\nThe following students play baseball, but not basketball:')
for name in baseball.difference(basketball):
    print(name)

print('\nThe following students play basketball, but not baseball:')
for name in basketball.difference(baseball):
    print(name)

print('\nThe following students play one sport, but not both:')
for name in baseball.symmetric_difference(basketball):
    print(name)
```

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Pse `pickle` për `birthdays`?

- Ruajtje **binar** e menjëhershme e fjalorit `birthdays` në disk.
- Rikthim i shpejtë i të dhënave midis ekzekutimeve pa formate të ndërmjetme.
- I përshtatshëm për **ruajtje lokale** (jo për shkëmbim publik).

Siguria

`pickle.load` **mos** e përdor me skedarë nga burime të pabesueshme mund të ekzekutojë kod gjatë deserializimit.

Integrimi minimal në birthdays2.py

```python
import pickle

DB_FILE = 'birthdays.db'  # emri i skedarit binar

def save_db(birthdays, filename=DB_FILE):
    with open(filename, 'wb') as f:
        pickle.dump(birthdays, f, protocol=pickle.HIGHEST_PROTOCOL)
    print(f'Saved {len(birthdays)} entries to {filename}')

def load_db(filename=DB_FILE):
    try:
        with open(filename, 'rb') as f:
            data = pickle.load(f)
        print(f'Loaded {len(data)} entries from {filename}')
        return data
    except FileNotFoundError:
        print('No DB found. Starting with an empty dictionary.')
        return {}
```

Menu e zgjeruar (Shto Save/Load)

```
LOOK_UP = 1; ADD = 2; CHANGE = 3; DELETE = 4; QUIT = 5; SAVE = 6; LOAD = 7

def get_menu_choice():
    print('\nFriends and Their Birthdays')
    print('---------------------------')
    print('1. Look up a birthday')
    print('2. Add a new birthday')
    print('3. Change a birthday')
    print('4. Delete a birthday')
    print('5. Quit the program')
    print('6. Save to disk (pickle)')
    print('7. Load from disk (pickle)\n')
    choice = int(input('Enter your choice: '))
    while choice < LOOK_UP or choice > LOAD:
        choice = int(input('Enter a valid choice: '))
    return choice
```

main() me ruajtje/ngarkim

```
def main():
    birthdays = {}              # ose: birthdays = load_db()
    choice = 0
    while choice != QUIT:
        choice = get_menu_choice()
        if choice == LOOK_UP:   look_up(birthdays)
        elif choice == ADD:     add(birthdays)
        elif choice == CHANGE:  change(birthdays)
        elif choice == DELETE:  delete(birthdays)
        elif choice == SAVE:    save_db(birthdays)
        elif choice == LOAD:    birthdays = load_db()
```

Opsional: Auto-Save

Thirr `save_db(birthdays)` pas `add`, `change`, `delete` nëse do ruajtje automatike.

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Pse `json` krahas `pickle`?

- **Tekst i lexueshëm**, i përshtatshëm për ndarje me të tjerët apo për ruajtje në Git.
- **Ndërgjuhësh**: mund të lexohet nga JavaScript, Java, etj.
- Më **i sigurt** për input publik (nuk ekzekuton kod).

Strategji

Përdor `pickle` për ruajtje lokale të shpejtë; `json` për eksport/import dhe versionim.

Funksione export_json / import_json I

```
import json

JSON_FILE = 'birthdays.json'

def export_json(birthdays, filename=JSON_FILE):
    with open(filename, 'w', encoding='utf-8') as f:
        json.dump(birthdays, f, ensure_ascii=False, indent=2, sort_keys=True)
    print(f'Exported {len(birthdays)} entries to {filename}')
```

Funksione export_json / import_json II

```
def import_json(filename=JSON_FILE):
    try:
        with open(filename, 'r', encoding='utf-8') as f:
            data = json.load(f)
        if not isinstance(data, dict):
            print('Invalid JSON format (expected an object/dict).'); return {}
        print(f'Imported {len(data)} entries from {filename}')
        return data
    except FileNotFoundError:
        print('JSON file not found.'); return {}
    except json.JSONDecodeError as e:
        print('Invalid JSON:', e); return {}
```

Menu e zgjeruar për JSON

```
EXPORT_JSON = 8
IMPORT_JSON = 9

def get_menu_choice():
    print('\nFriends and Their Birthdays')
    print('---------------------------')
    print('1. Look up a birthday')
    print('2. Add a new birthday')
    print('3. Change a birthday')
    print('4. Delete a birthday')
    print('5. Quit the program')
    print('6. Save to disk (pickle)')
    print('7. Load from disk (pickle)')
    print('8. Export to JSON')
    print('9. Import from JSON\n')
    choice = int(input('Enter your choice: '))
    while choice < LOOK_UP or choice > IMPORT_JSON:
        choice = int(input('Enter a valid choice: '))
    return choice
```

main() me JSON

```
def main():
    birthdays = {}
    choice = 0
    while choice != QUIT:
        choice = get_menu_choice()
        if choice == LOOK_UP:          look_up(birthdays)
        elif choice == ADD:            add(birthdays)
        elif choice == CHANGE:         change(birthdays)
        elif choice == DELETE:         delete(birthdays)
        elif choice == SAVE:           save_db(birthdays)
        elif choice == LOAD:           birthdays = load_db()
        elif choice == EXPORT_JSON:    export_json(birthdays)
        elif choice == IMPORT_JSON:    birthdays = import_json()
```

Këshillë

Për datat, mbaji si `string` (`"dd/mm/yyyy"`) dhe **valido** me `datetime.strptime` para ruajtjes.

Suksese në laborator!

Pyetje?